

16) Develop a Dynamic To-Do List Application

HTML input controls, CSS list styling, JavaScript arrays, objects, DOM manipulation, event listeners and add, edit and delete functions.

Code

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Dynamic To-Do List</title>
```

```
<style>
```

```
  * {
```

```
    box-sizing: border-box;
```

```
    margin: 0;
```

```
    padding: 0;
```

```
  }
```

```
  body {
```

```
    font-family: Arial, sans-serif;
```

```
    background: linear-gradient(135deg, #667eea, #764ba2);
```

```
    min-height: 100vh;
```

```
    display: flex;
```

```
    justify-content: center;
```

```
    align-items: center;
```

```
    padding: 20px;
```

```
}
```

```
.container {  
  width: 100%;  
  max-width: 550px;  
  background: white;  
  padding: 30px;  
  border-radius: 18px;  
  box-shadow: 0 10px 30px rgba(0, 0, 0, 0.25);  
}
```

```
h1 {  
  text-align: center;  
  color: #4a3f78;  
  margin-bottom: 25px;  
}
```

```
.input-box {  
  display: flex;  
  gap: 10px;  
  margin-bottom: 25px;  
}
```

```
#taskInput {  
  flex: 1;  
  padding: 13px;  
  border: 2px solid #ddd;  
  border-radius: 8px;
```

```
font-size: 16px;  
outline: none;  
}
```

```
#taskInput:focus {  
    border-color: #667eea;  
}
```

```
#addBtn {  
    padding: 13px 20px;  
    border: none;  
    border-radius: 8px;  
    background: #667eea;  
    color: white;  
    cursor: pointer;  
    font-size: 16px;  
}
```

```
#addBtn:hover {  
    background: #4f63c4;  
}
```

```
ul {  
    list-style: none;  
}
```

```
li {  
    display: flex;
```

```
    align-items: center;
    justify-content: space-between;
    gap: 10px;
    background: #f5f5f5;
    padding: 12px;
    margin-bottom: 10px;
    border-radius: 8px;
    border-left: 5px solid #667eea;
}
```

```
.task-text {
    flex: 1;
    word-break: break-word;
}
```

```
.actions {
    display: flex;
    gap: 7px;
}
```

```
.edit,
.delete {
    border: none;
    padding: 8px 12px;
    border-radius: 6px;
    color: white;
    cursor: pointer;
}
```

```
.edit {  
  background: #f4a261;  
}
```

```
.delete {  
  background: #e63946;  
}
```

```
.edit:hover {  
  background: #d88945;  
}
```

```
.delete:hover {  
  background: #c72f3b;  
}
```

```
.empty {  
  text-align: center;  
  color: #888;  
  padding: 20px;  
}
```

```
@media (max-width: 500px) {  
  .container {  
    padding: 20px;  
  }  
}
```

```
.input-box {
    flex-direction: column;
}

#addBtn {
    width: 100%;
}

li {
    align-items: flex-start;
    flex-direction: column;
}

.actions {
    width: 100%;
}

.edit,
.delete {
    flex: 1;
}

}

</style>

</head>

<body>

<div class="container">
```

```
<h1> My To-Do List</h1>
```

```
<div class="input-box">
```

```
  <input type="text" id="taskInput" placeholder="Enter a task">
```

```
  <button id="addBtn">Add Task</button>
```

```
</div>
```

```
<ul id="taskList"></ul>
```

```
</div>
```

```
<script>
```

```
  let tasks = [];
```

```
  const taskInput = document.getElementById("taskInput");
```

```
  const addBtn = document.getElementById("addBtn");
```

```
  const taskList = document.getElementById("taskList");
```

```
  function displayTasks() {
```

```
    taskList.innerHTML = "";
```

```
    if (tasks.length === 0) {
```

```
      taskList.innerHTML = '<div class="empty">No tasks added yet</div>';
```

```
      return;
```

```
    }
```

```
    tasks.forEach(function(task, index) {
```

```
      const li = document.createElement("li");
```

```
const span = document.createElement("span");
```

```
span.className = "task-text";
```

```
span.textContent = task.name;
```

```
const actions = document.createElement("div");
```

```
actions.className = "actions";
```

```
const editBtn = document.createElement("button");
```

```
editBtn.className = "edit";
```

```
editBtn.textContent = "Edit";
```

```
const deleteBtn = document.createElement("button");
```

```
deleteBtn.className = "delete";
```

```
deleteBtn.textContent = "Delete";
```

```
editBtn.addEventListener("click", function() {
```

```
    editTask(index);
```

```
});
```

```
deleteBtn.addEventListener("click", function() {
```

```
    deleteTask(index);
```

```
});
```

```
actions.appendChild(editBtn);
```

```
actions.appendChild(deleteBtn);
```

```
li.appendChild(span);
```

```
li.appendChild(actions);
```

```
        taskList.appendChild(li);  
    });  
}
```

```
function addTask() {  
    const taskName = taskInput.value.trim();  
  
    if (taskName === "") {  
        alert("Please enter a task");  
        return;  
    }  
}
```

```
const task = {  
    id: Date.now(),  
    name: taskName  
};
```

```
tasks.push(task);  
taskInput.value = "";  
displayTasks();  
}
```

```
function editTask(index) {  
    const newTask = prompt("Edit task:", tasks[index].name);  
  
    if (newTask !== null && newTask.trim() !== "") {  
        tasks[index].name = newTask.trim();  
    }  
}
```

```
        displayTasks();
    }
}
```

```
function deleteTask(index) {
    if (confirm("Are you sure you want to delete this task?")) {
        tasks.splice(index, 1);
        displayTasks();
    }
}
```

```
addBtn.addEventListener("click", addTask);
```

```
taskInput.addEventListener("keypress", function(event) {
    if (event.key === "Enter") {
        addTask();
    }
});
```

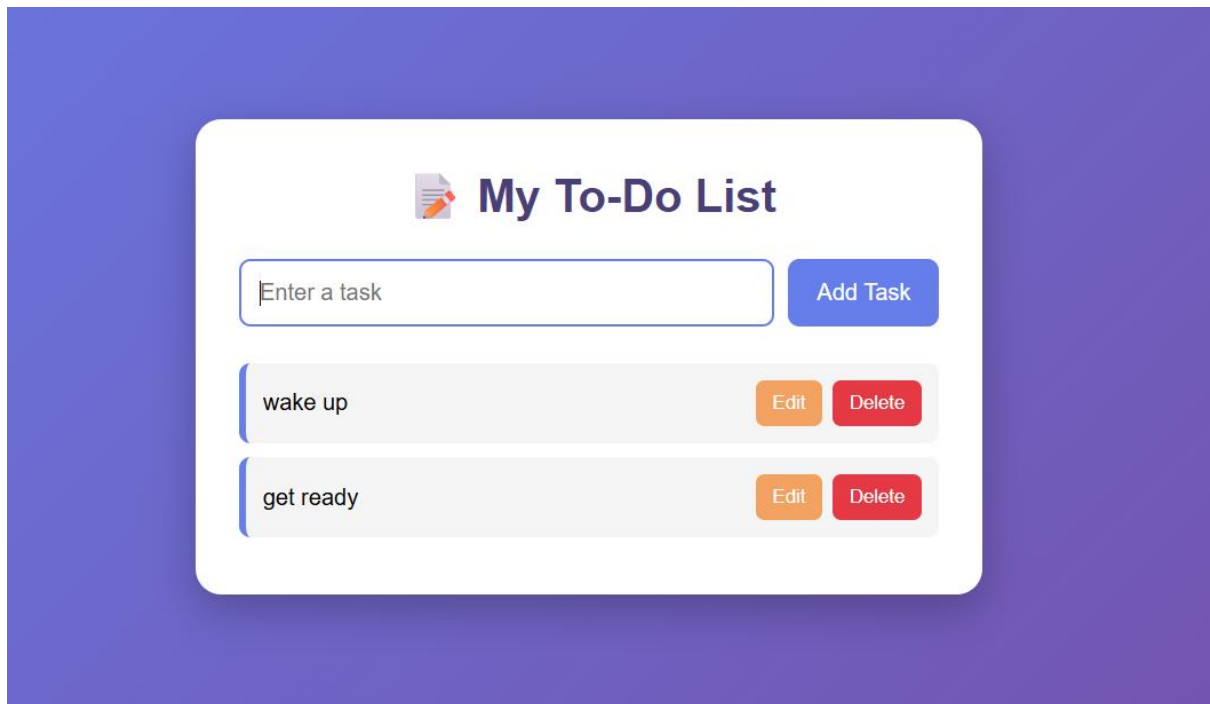
```
displayTasks();
```

```
</script>
```

```
</body>
```

```
</html>
```

Output



The image shows a web application titled "My To-Do List". It features a white rounded rectangle centered on a solid purple background. At the top of the white area is a header with a notepad icon and the title "My To-Do List". Below the header is a form with a text input field containing the placeholder text "Enter a task" and a blue "Add Task" button. Underneath the form are two task entries, each in a light gray box. The first entry is "wake up" with an orange "Edit" button and a red "Delete" button. The second entry is "get ready" with an orange "Edit" button and a red "Delete" button.

My To-Do List

Add Task

wake up Edit Delete

get ready Edit Delete